

Problem_1: The following table gives the data on the distribution of income of urban households in a certain country during a certain year.

Income Class (taka)	Percentage of household
Under 500	13.6
500-999	28.9
1000-1999	32.5
2000-2999	10.6
3000-3999	5.6
4000-4999	3.1
5000-5999	1.7
6000-7999	1.7
8000-9999	0.7
10000-14999	0.8
15000-24999	0.5
25000 or over	0.3

Fit a Pareto distribution to the data. With the estimated values of the parameters derive the concentration ratio.

Solution:

The Pareto distribution of income is given by $y = Ax^{-\gamma}$ (1)

where y = number of households having income x or more

x = income; A and γ are unknown parameters.

Taking logarithms of both sides of (1) we get

$$\log_{10}y = \log_{10}A - \gamma\log_{10}x \quad (2)$$

(2) can be written as $Y = \beta_0 + \beta_1X$ (3)

where $Y = \log_{10}y$, $\beta_0 = \log_{10}A$, $\beta_1 = -\gamma$, $X = \log_{10}x$

Adding an error term u to equation (3), we have

$$Y = \beta_0 + \beta_1X + u \quad (4) ; u \sim N(0, \sigma^2)$$

OLS estimates of β_0 and β_1 are, $\hat{\beta}_1 = \dots\dots\dots$ and $\hat{\beta}_0 = \dots\dots\dots$

The estimates of parameters A and γ are

$$\hat{A} = \text{Anti log}_{10} \hat{\beta}_0 ; \quad \hat{\gamma} = -\hat{\beta}_1$$

The estimated Pareto distribution is $y = \hat{A}x^{-\hat{\gamma}}$

In the given data, the lowest income is unknown. We set the lowest income arbitrarily at $x_0 = 100$ taka (say).

The table showing the calculation for estimating the Pareto distribution.

Income (x)	Percentage of households having income $\geq x$ (y)	$X = \log_{10} x$	$Y = \log_{10} y$
100	100.0		
500	86.4		
1000	57.5		
2000	25.0		
3000	14.4		
4000	8.8		
5000	5.7		
6000	4.0		
8000	2.3		
10000	1.6		
15000	0.8		
25000	0.3		

Here, n = number of classes = 12.

$$\text{Lorenz concentration ratio } \hat{\rho} = \frac{1}{2\hat{\gamma}-1}$$

Now, $\hat{\beta}_1 = \text{??????}$ and $\hat{\beta}_0 = \text{??????}$

The estimates of parameters A and γ are, $\hat{A} = \text{Anti log}_{10} \hat{\beta}_0 = \text{??????}$; $\hat{\gamma} = -\hat{\beta}_1 = \text{??????}$

The estimated Pareto distribution is $y = (\text{??????})x^{-(\text{??????})}$

$$\text{Lorenz concentration ratio } \hat{\rho} = \frac{1}{2\hat{\gamma}-1} = \text{??????}$$

Problem_2: Fit a lognormal distribution to the following data and test the goodness of fit.

Class interval of weekly wage (taka)	No. of workers in percentage
10-14	3.30
15-19	9.39
20-24	13.08
25-29	16.41
30-34	13.87
35-39	11.89
40-44	9.30
45-49	6.79
50-54	5.82
55-59	3.39
60 and above	6.76

Solution: The form of lognormal distribution is

$$f(x) = \frac{1}{x \cdot \sigma \cdot \sqrt{2 \cdot \pi}} \cdot \exp \left[-\frac{(\ln x - \mu)^2}{2 \cdot \sigma^2} \right], 0 < x < \infty.$$

The last class interval is open, we set the upper limit arbitrarily at 64.

Class interval	Mid values (X_i)	No. of workers in percentage (f_i)	$f_i X_i$	$f_i X_i^2$
10-14		3.30		
15-19		9.39		
20-24		13.08		
25-29		16.41		
30-34		13.87		
35-39		11.89		
40-44		9.30		
45-49		6.79		
50-54		5.82		
55-59		3.39		
60-64		6.76		

Now, the sample mean and variance are $\bar{X} = \frac{\sum f_i X_i}{N} = \text{??????}$,where $\sum f_i = N$

$$S^2 = \frac{1}{N-1} \left[\sum f_i X_i^2 - \frac{(\sum f_i X_i)^2}{N} \right] = \text{?????}$$

We assume that the given sample data have come from a lognormal population with unknown parameters μ and σ^2 . The estimates of μ and σ^2 , using method of moments are given by

$$\hat{\mu} = \ln \left[\frac{\bar{x}^2}{\sqrt{S^2 + \bar{x}^2}} \right] = \text{?????}$$

$$\hat{\sigma}^2 = \ln \left[\frac{S^2 + \bar{x}^2}{\bar{x}^2} \right] = \text{?????}$$

Then the fitted lognormal distribution is

Class interval	Observed frequency (O _i)	Class boundary	In of class upper boundary (y _i)	$Z_i = \frac{y_i - \hat{\mu}}{\sigma}$	$\phi(Z_i)$	$\Delta\phi(Z_i)$	Expected frequency [E _i =N* $\Delta\phi(Z_i)$]
		Below 9.5		-3.2413	=NORM.S.DIST(-3.2413,1)=0.0006	-	-
10-14	3.30	9.5-14.5					
15-19	9.39	14.5-19.5					
20-24	13.08	19.5-24.5					
25-19	16.41	24.5-29.5					
30-34	13.87	29.5-34.5					
35-39	11.89	34.5-39.5					
40-44	9.30	39.5-44.5					
45-49	6.79	44.5-49.5					
50-54	5.82	49.5-54.5					
55-59	3.39	54.5-59.5					
60-64	6.76	59.5-64.5					

H₀: The sample data have come from lognormal distribution with mean=??? and standard deviation=????

H₁: H₀ is not true.

Test statistic is $\chi^2 = \sum_{i=1}^k \frac{O_i^2}{E_i^2} - N \sim \chi_{k-p-1}^2$; where k=no. of classes=11 and p=no. of parameters=2. or

Test statistic is $\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi_{k-p-1}^2$; where k=no. of classes=11 and p=no. of parameters=2.

****Tabulated value of χ^2 (from excel) = CHISQ.INV.RT(0.05,8) =15.5**

Comment: